

Linear Aeroelastic Modelling for Stability Analysis and Structural Sizing

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Abstract—Compliance with aeroelastic requirements is necessary for aircraft certification and must hence be considered in the design. The interaction between aircraft structures and the surrounding air can be complex and demands very careful modelling. This paper presents a linear aeroelastic model coupling engineering beam theory and potential flow theory, thereby demonstrating the basic implementation of a simpler aeroelasticity model. The model is experimentally verified and used to predict the onset of divergence, control reversal and flutter instabilities and to size the wing structure of a small utility aircraft according to stability considerations and loads at quasi-steady flight manoeuvres.

Index Terms—Aeroelasticity, stability, divergence, control reversal, flutter, FEM, structural optimisation.

I. INTRODUCTION

AIRCRAFT structures, in particular wings, may exhibit significant elastic deformation during flight when subjected to aerodynamic loads. The interaction between the fluid and structure needs to be taken into account when designing aircraft as it may cause static or dynamic stability problems or affect controllability, potentially causing structural failure or severe reduction of handling qualities. Certification specifications such as the CS-23 [1] therefore define strict requirements for undesired aeroelastic phenomena to ensure safe flight within the flight envelope.

Aeroelastic models can help to size structural elements, investigate the static and dynamic stability and predict the influence on flight dynamics [2]. This is of particular interest for the assessment and development of new aircraft configurations and topic of ongoing research. The *strut-braced wing* and the *UAV MALE* (fig. 1) are aircraft configurations for which aeroelastic models have been applied in the recently finished the European research programme *AGILE* [3]. Both studies aimed to achieve structural optimisation in the preliminary design phase of the aircraft [4], [5].

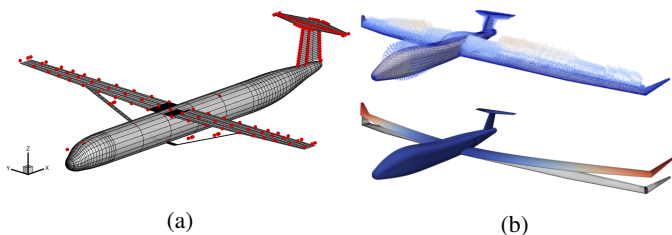


Figure 1: (a) Strut-braced wing aerodynamic model [4] and (b) UAV MALE with aerodynamic loads and displacement [5]

This paper aims to demonstrate a basic aeroelastic modelling procedure for an aircraft wing. A linear finite element (FE) beam model is used to model the wing structure. Aerodynamic loads are computed using potential flow theory. First, the general modelling approach will be presented, followed by an experimental validation. Finally, the model is used to size the wing structure of the *DHC-6 Twin Otter* aircraft (fig. 8, p. 5). Some *aeroelastic* stability problems will be addressed here, namely *divergence*, *control reversal* and *classical flutter*. These concepts are briefly explained below.

A. Divergence

Aeroelastic divergence is a static instability that may occur if a lifting surface does not have the required stiffness to sustain the aerodynamic loads. At divergence the deformation of the wing is continuously amplified as the aerodynamic loading increases which contributes to even further structural deformation. When the elastic restoring forces are overcome, a point of structural failure is reached. Main factors influencing this phenomenon are the dynamic pressure and wing torsional stiffness. Forward swept wings are particularly sensitive to divergence as the forward sweep introduces a disadvantageous coupling between bending and torsion deformations [6].

B. Control reversal

Control reversal is another static instability which may occur due to insufficient structural stiffness. When flight control surfaces such as ailerons are deflected, they usually introduce forces into the wing at some distance away from its elastic axis which causes the wing to twist accordingly. This fluid-structure interaction can subsequently reduce the effectiveness of control surface deflections or even reverse their intended effects.

C. Flutter

Unlike divergence and control reversal, flutter is a *dynamic* instability in which unsteady aerodynamic loads continuously feed energy into the structure. Structural mass distributed along the wing span is accelerated in an oscillating motion. Hence, not only elastic properties but also the mass distribution influences the response of the wing. Different types of flutter can be distinguished such as *classical flutter*, *stall flutter*, *transonic flutter* and *buffeting* [7].

II. THEORY AND MODELLING

A. General approach of the aeroelastic modelling

One main objective of the presented aeroelastic model is to investigate the static and dynamic response of a wing structure when being subjected to aerodynamic loads. In particular, the model should predict the onset of *critical conditions* for divergence, control reversal and flutter. The general approach to this problem involves solving the equations of motion of the structure. Structural damping will not be considered and only aerodynamic loads will be taken into account as an external source for excitations and deformations of the structure. The *time-domain* equations of motion for a discretised structure can be written in matrix form as

$$\mathbf{M} \ddot{\mathbf{v}} + \mathbf{K} \mathbf{v} = \mathbf{f}_a(t, \dot{\mathbf{v}}, \ddot{\mathbf{v}}, \dots) \quad (1)$$

where $\mathbf{M}$ is a mass matrix, $\mathbf{K}$ a stiffness matrix, $\mathbf{v}$ the displacement field, $\mathbf{f}_a$ the aerodynamic load vector and t time. Note that other external loads such as gust loads can be taken into account in a similar way. Flutter analysis, which involves unsteady aerodynamic loads, may be performed in the *frequency-domain*. The frequency-domain formulation of eq. (1) is obtained after a Laplace transform:

$$\mathbf{M} p^2 \hat{\mathbf{v}} + \mathbf{K} \hat{\mathbf{v}} = \hat{\mathbf{f}}_a(p) = q \mathbf{Q}(p) \hat{\mathbf{v}} \quad (2)$$

Here, $p = \sigma + i\omega$ is the Laplace variable with the damping σ and the angular frequency ω . q denotes dynamic pressure, $\mathbf{Q}(p)$ is an aerodynamic transfer matrix and $\hat{\mathbf{v}}$ a complex displacement vector. The aerodynamic matrix $\mathbf{Q}$ is thus formulated such that a multiplication with dynamic pressure and displacement vector ($q \mathbf{Q} \hat{\mathbf{v}}$) yields the discretised aerodynamic load.

B. Structural modelling

The wing is assumed to be a long and slender structure that with sufficient accuracy can be described with a beam model. This is a fairly common approach used for aeroelasticity models [2]. For the aeroelastic stability problems of interest (divergence, reversal and flutter), the “flapping” and twisting dofs are typically the most relevant ones. Other deformation modes are therefore neglected. Assuming linear Euler-Bernoulli bending and St. Venant torsion theory, the equations of motion of a beam are

$$m_y \ddot{w} - m_y s \ddot{\Theta} + (EI w'')'' = L_y(y, t) \quad (3)$$

$$J_y \ddot{\Theta} - m_y s \ddot{w} - (GK \Theta')' = M_y(y, t) \quad (4)$$

where m_y is the beam mass per unit span, J_y the rotational inertia per unit span and s the distance between the elastic axis and the centre of mass. EI and GK are bending and torsional stiffness, respectively. The external loads are aerodynamic lift L_y and pitch moment M_y , both given with respect to the elastic axis and as load per unit span. The beam equations (3) and (4) are partial differential equations for the deflection $w(y, t)$ and the twist $\Theta(y, t)$, coupled by the parameter s .

As shown by Borglund and Eller [7], the beam model can be written in a discretised FE formulation in the form of eq. (1). To describe wing bending and torsion with the FE model, three

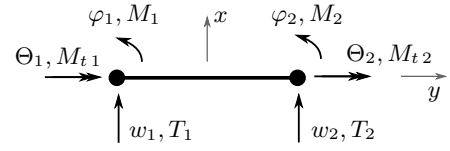


Figure 2: FE beam element with 3 dof per node

dofs per node j are required, namely the deflection w_j , the beam angle φ_j and the twist Θ_j (fig. 2). Thus, the global nodal displacement vector for a FE model with N elements becomes

$$\mathbf{v}(t) = [w_1(t), \varphi_1(t), \Theta_1(t), w_2(t), \varphi_2(t), \Theta_2(t), \dots, w_{N+1}(t), \varphi_{N+1}(t), \Theta_{N+1}(t)]^T$$

C. Aerodynamic modelling

To model the aerodynamic loads on the wing structure a two-dimensional (2D) linear, unsteady potential flow model is chosen. As the aeroelasticity model is restricted to slender wing structures it is assumed that the flow is approximately two-dimensional at each spanwise position. Furthermore, only low speed conditions are being analysed, thus allowing the simplification of incompressible flow with reasonable accuracy.

Strip theory [7] is deemed to be sufficient for the model. The aerodynamic lift $L_y(y, t)$ and pitch moment $M_y(y, t)$ (per unit span) are now introduced to eqs. (3) and (4).

$$L_y = q c \left(\frac{L_h}{b} h + L_\alpha \alpha + L_\delta \delta \right) \quad (5)$$

$$M_y = q c^2 \left(\frac{M_h}{b} h + M_\alpha \alpha + M_\delta \delta \right) \quad (6)$$

Here, q denotes dynamic pressure, c chord length and b the semi-chord. The variables L_j and M_j , where $j = h, \alpha, \delta$, are lift and moment coefficients describing the influence from elastic translational and rotational deformations as well as from the control surface deflection. Analytical expressions for these coefficients (including frequency-domain formulations) have been derived by Theodorsen as summarised in [7].

D. Solutions

The analyses of divergence, reversal and flutter can all be reduced to eigenvalue problems, which shall be briefly mentioned here. Derivations and more detailed solution procedures are given by Borglund and Eller [7].

1) *Divergence analysis*: The condition for the onset of divergence is the point at which the structure is (still) at *static* equilibrium between aerodynamic loads and the elastic reaction force, but where an infinitesimal increase of loading would lead to failure (or rather to an infinite deformation as a linear structure model is used). This condition can be traced back to an eigenvalue problem.

$$[\mathbf{K} - q \mathbf{Q}_0] \mathbf{v} = \mathbf{0} \quad (7)$$

The smallest real eigenvalue q is the divergence dynamic pressure q_D and the corresponding eigenvector is $\mathbf{v}_D$. The matrix $\mathbf{Q}_0 := \mathbf{Q}(k = 0)$ is also referred to as steady-state aerodynamic matrix which has been computed for a reduced frequency of zero (see section II-D3).

2) *Control reversal analysis*: The control reversal problem can be analysed in a similar way. The condition is a static equilibrium at which the rolling moment coefficient $C_R = 0$, i.e. at reversal speed an aileron deflection does not produce any rolling moment. The corresponding eigenvalue problem is

$$[\mathbf{K} - q (\mathbf{Q}_0 + \mathbf{Q}_\delta)] \mathbf{v} = 0 \quad (8)$$

where $\mathbf{Q}_\delta$ describes the load contribution from the control surface deflection. The smallest real eigenvalue q is the reversal dynamic pressure q_R and the corresponding eigenvector is $\mathbf{v}_R$.

3) *Flutter analysis*: The flutter problem is analysed in the frequency-domain. From eq. (2) a nonlinear eigenvalue problem can be derived. It can be solved for the eigenvalue p and the complex-valued eigenvector $\hat{\mathbf{v}}$.

$$[\mathbf{M} p^2 + \mathbf{K} - q \mathbf{Q}(\hat{p})] \hat{\mathbf{v}} = 0 \quad (9)$$

Here $\hat{p}$ is the *reduced* Laplace variable

$$\hat{p} = p \frac{b}{u} = \frac{\sigma b}{u} + i \frac{\omega b}{u} = \gamma + i k \quad (10)$$

where γ is a non-dimensional damping and $k = \omega b/u$ is the reduced frequency of vibration. The reference length b is the wing semi-chord and u is the flow velocity. k is a dimensionless quantity and can be interpreted as a degree of unsteadiness. The eigenvalue problem eq. (9) is approximated since the aerodynamic model provides frequency-domain data $\mathbf{Q}(k)$ and its solution involves an iterative procedure, the so-called p-k method, which is described in [7].

III. EXPERIMENTAL MODEL VERIFICATION

The aeroelasticity model outlined in section II was implemented using *Matlab* and experimentally verified using a *flat plate wing model*. The model and its geometry are described in more detail in the appendix A. First, free vibrations were measured in a ground vibration test (gvt). Results obtained from this test were used to fine-tune stiffness and mass data. In a following wind tunnel experiment the critical speeds for divergence, aileron reversal and flutter were measured and compared to predictions made by the computational model.

A. Material parameters

Besides correctly modelling the wing geometry with the FE model it is also crucial to determine the material parameters accurately as they influence the mass and stiffness matrices. In a first step, base line material parameters were determined from a smaller, rectangular plate sample (same material as used in wind tunnel model). The plate density ρ was estimated by measuring the geometry and the mass of the sample. Preliminary values for the Young's modulus E and shear modulus G were obtained from simply loading the sample in bending and torsion and measuring the corresponding deformations. Assuming linear Euler-Bernoulli bending and St. Venant torsion theory, E and G could be estimated.

In a second step, the base line material data was used to make predictions for eigenfrequencies in bending and torsion. Two different structures were modelled and then tested in a gvt: firstly, the clamped wind tunnel model as shown in

fig. 15 (p. 9) and secondly a free-hanging¹ “clean” plate (no ailerons and no joints) with the same overall dimensions like the wind tunnel model. Additional masses were attached to the latter structure and changes in eigenfrequencies could be compared to results from the computational model. Bending and torsional vibration modes of the structures were excited manually, either by bending or twisting the plates by hand or by applying a more discrete impulse with an excitation hammer. The structural response was measured with an accelerometer; the measurements could directly be post-processed with a fast fourier transform (FFT). In case of the clamped wind tunnel model an additional optical system (four cameras with infrared light sources) was used which allowed to measure time-dependent coordinates of three reflective markers attached to the wing tip. These marker positions could be processed with a FFT to obtain vibration frequencies.

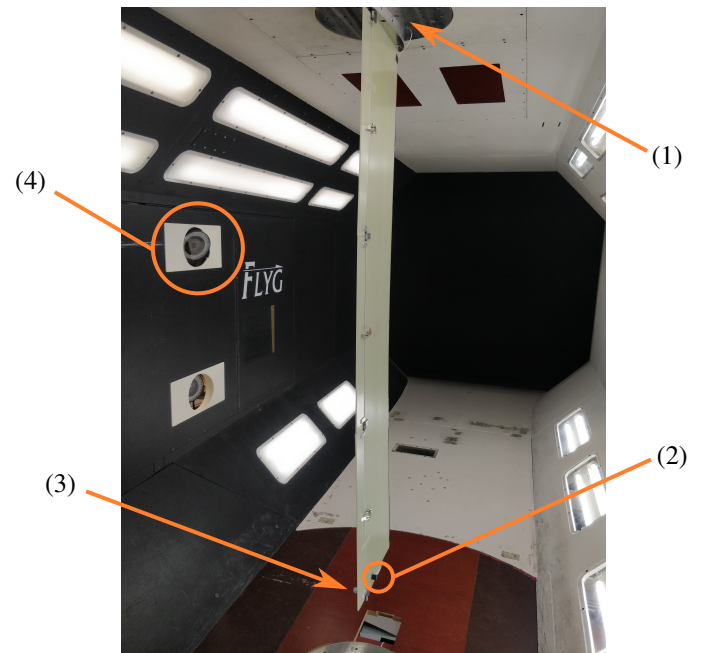


Figure 3: Wing model mounted in the wind tunnel: (1) clamp, (2) accelerometer, (3) reflective marker and (4) camera

The experimentally obtained FFT results were used to fine-tune the base-line values of E , G and ρ so that the computational model showed good agreement with experimental data. The determined material data is summarised in table V (p. 9) and the measured and calculated first seven eigenfrequencies are compared in table I. Generally, good agreement between the FE model and experiment was achieved. However, the fifth vibration mode as predicted by the FE model was neither found with the accelerometer nor with optical sensors. The optical measurements were good at determining the lowest frequencies (at low structural accelerations), but the accelerometer proved to pick up the higher frequencies more reliably.

¹With respect to the 3 dof FE model, the free-hanging plate can be considered to be unconstrained.

Table I: Predicted and measured free vibration frequencies of the wind tunnel model

Method	f_1 [Hz]	f_2 [Hz]	f_3 [Hz]	f_4 [Hz]	f_5 [Hz]	f_6 [Hz]	f_7 [Hz]
Theory (FE model)	0.92	5.06	5.77	14.52	16.51	25.62	29.77
Experiment (accelerometer)	—	5.29	5.50	14.51	—	25.65	31.64
Experiment (optical system)	0.98	5.30	5.56	14.35	—	—	—

B. Stability experiments

Once material data was calibrated to fit the gvt results, reasonable predictions for divergence, reversal and flutter could be made. As described in section II-C strip theory was used to model the aerodynamic loads. Additionally, frequency-domain aerodynamic data based on doublet-lattice method (DLM) for the flat plate geometry was provided externally and also used to predict the aeroelastic instabilities.

The wing model was clamped at the wing root and hung vertically from the ceiling of the wind tunnel to keep the influence from gravity as small as possible (fig. 3). Critical speeds for divergence, reversal and flutter were determined by visual inspection of the model during the test. Frequencies of vibrations were again measured with an accelerometer and with the optical system as for the gvt. The predictions for the onset of instabilities based on strip theory and DLM aerodynamics are summarised in table II together with the experimental results.²

Table II: Predicted and measured critical speeds

Instability	Parameter	Strip	DLM	Experiment
Divergence	u [m/s]	20.6	25.4	23.5
Reversal	u [m/s]	18.9	22.4	≥ 21
Flutter	u [m/s]	15.3	17.2	18.8
Flutter	f [Hz]	2.91	3.14	3.21

Flutter was predicted to have the lowest critical airspeed and could successfully be demonstrated, though at a higher airspeed than suggested by the computational model. For the flutter demonstration airspeed was very slowly increased and discrete disturbances were added to the wing tip by pushing it with a stick. Until the reported critical speed oscillations that followed this disturbance decreased indicating stable behaviour. At the critical speed disturbances from the airflow (turbulent flow) were sufficient to cause an unstable structural response, i. e. flutter. In figs. 4 to 6 damping and frequency behaviour of the critical mode are summarised for the strip theory and DLM aerodynamic models. In addition, fig. 5 shows the frequency change of the critical mode as obtained from the experiment.

As *aileron reversal* and *divergence* had higher critical speeds (table II), the wing was “mass-balanced” so that to increase the flutter speed.³ In a first mass-balancing attempt using 600 g⁴ divergence could be demonstrated. In this case strip theory again underestimated the critical airspeed, but DLM overestimated the speed (table II).

²Note that all computed airspeeds in this report are given with respect to an air density of 1.225 kg/m³.

³Theory suggests that the mass distribution only influences flutter, but not divergence and reversal as can be seen from eqs. (7) to (9).

⁴The following additional masses were attached at the wing’s leading edge: 100 g at $y = 1.1$ m, 200 g at $y = 1.3$ m and 300 g at $y = 1.6$ m.

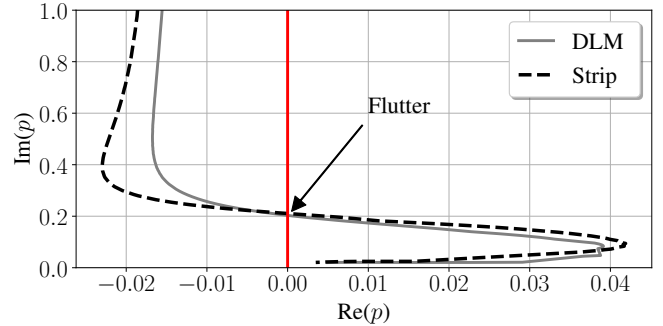
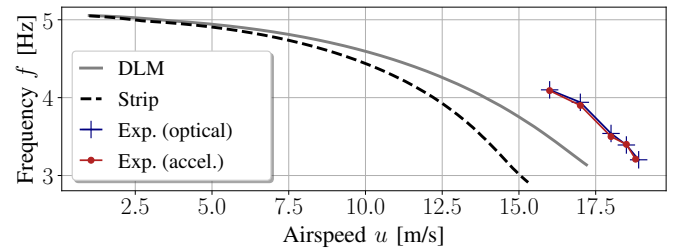


Figure 4: Root-locus plot for the wind tunnel model

Figure 5: Frequency change of critical mode with airspeed until onset of flutter (*exp.*: experiment, *accel.*: accelerometer)

In order to induce *aileron reversal*, all three ailerons were deflected to approximately 5°. In a first test run using the same additional masses as for the divergence test, the wing deflected and twisted as expected with increasing airspeed. However, at approximately 21 m/s pronounced flutter began so that the test had to be aborted. In a second mass balancing attempt using 500 g⁵ the wing deflected and twisted as before. Again, at 21 m/s a flutter instability was encountered so that the test had to be aborted.

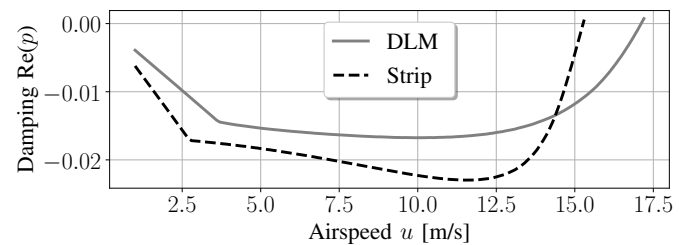


Figure 6: Aerodynamic damping of critical mode with airspeed until onset of flutter

⁵The following additional masses were attached at the wing’s leading edge: 200 g at $y = 1.3$ m and 300 g at $y = 1.6$ m.

The measured wing tip deflection for this second test run and the corresponding airspeed and time are shown in fig. 7. A maximum wing tip deflection was reached at 18.5 m/s, but it then decreased towards the point at which the wing began to flutter. The exact aileron reversal speed could thus not be measured but it could be determined to be equal or greater than 21 m/s.

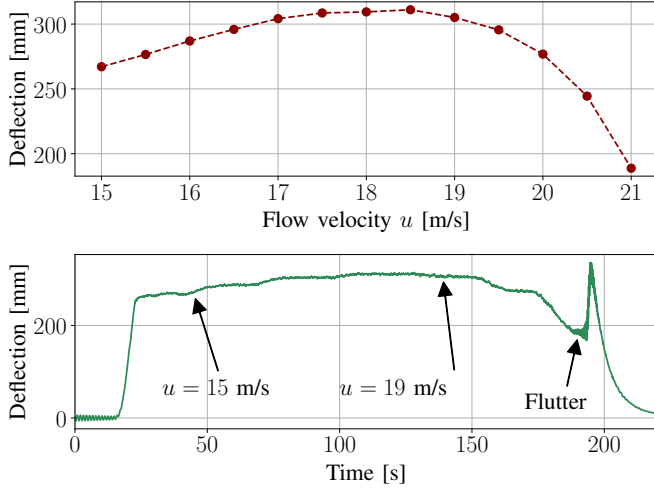


Figure 7: Wing tip deflection during the reversal experiment (measured with the optical system with respect to a reference point; presented data was averaged from three marker positions)

C. Discussion

Overall, the DLM aerodynamics provided closer predictions of the critical speeds than strip theory. As DLM provides a more accurate aerodynamic model, also taking spanwise effects into account, its better predictions seem reasonable. While all strip theory estimates were conservative, the DLM provided a non-conservative estimate of the divergence speed. Both theories correctly reflected a frequency decrease when approaching the critical speed (fig. 5).

It should also be noted that the wind tunnel model proved to be a fairly sensitive model. At the start of the stability experiment, a hinge was slightly loose. This caused significant oscillations at very low airspeeds (< 12 m/s). Furthermore, gravity might have had a relevant influence as well. As the wing was deflected during the aileron reversal experiment, gravity will induce loads on the structure. However, these were not modelled and its influence was not investigated.

IV. SIZING OF THE DHC-6 TWIN OTTER WING

A. General objective

The developed aeroelastic model has been applied to size the wing structure of the *DHC-6 Twin Otter*. The Twin Otter is an all-metal aircraft in a high wing monoplane configuration manufactured by *Viking Air Limited* in Canada. It is powered by two wing-mounted turboprop engines and can carry a pilot, co-pilot, and up to 19 passengers (Series 400) [8].⁶

⁶For the following analysis the aircraft is treated according to the utility category with a maximum take-off weight (MTOW) of 5670 kg (CS 23.1 [1]).

The sizing of the wing structure was based on considerations of *quasi-static* manoeuvres at the limit load factors $n_z = +4.4$ and $n_z = -1.8$. Furthermore, the wing was designed such that most basic requirements defined in CS-23.629 [1] for divergence, flutter and reversal would be fulfilled. Finally, the wing structure was optimised to achieve minimal structural weight while complying with these requirements.

B. Modelling, general assumptions and simplifications

As indicated in fig. 8, only the main wing was modelled as an elastic structure using a FE beam model. Other structural components were assumed to be stiff and only considered with their inertial or aerodynamic effects on the wing structure. A summary of geometric and inertial data used for this analysis is provided in section B-A. In order to capture potential asymmetric deformation modes the *full* wing was modelled. After performing convergence studies on the critical speeds for divergence, reversal and flutter as well as for the stress analysis, 100 finite elements were deemed to yield sufficient numerical accuracy and used for all subsequent analyses.

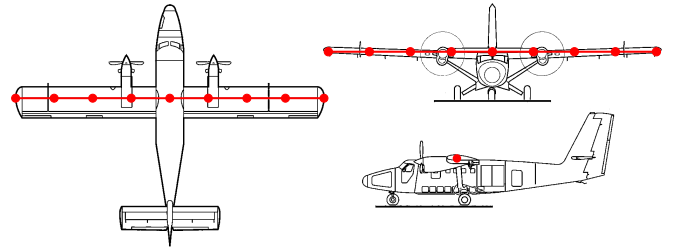


Figure 8: General arrangement of the DHC-6 Twin Otter, modified from [8]. Location of FE model is indicated in red.

The *wing structure* was modelled as a fully rectangular wing with two load-bearing elements: A closed wing box (indicated in blue in fig. 9) was assumed to contribute to bending stiffness and to carry all torsional loads, and a wing spar (indicated in green in fig. 9) was assumed to carry all normal stresses due to bending. The elastic axis was assumed to be located at 35 % chord length (wing spar position). Shear stresses due to transverse loads were assumed to be carried by the web of the wing spar and the skin, but not considered any further in this analysis. The struts as can be seen in fig. 8 can stiffen the wing and therefore potentially help to reduce structural weight (at the cost of aerodynamic efficiency). However, the struts were not modelled in the presented work. The structural support was modelled in a highly simplified manner as only the wing root was clamped in a single node.

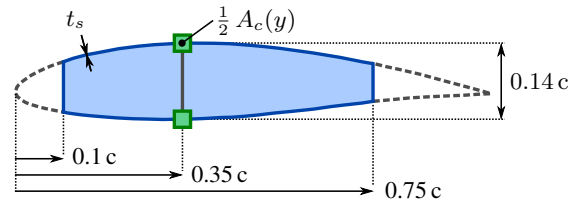


Figure 9: Simplified wing structure with load-bearing elements (wing skin and spar) coloured

The *aileron*s were assumed to span the outer half of the semi-span and to have a chord length of 20 % of the wing chord.

Inertia contributions (mass and rotational of inertia) of large components such as engines, fuselage and empennage were taken into account (details in section B-A). The total aircraft mass was kept constant at MTOW (5670 kg) for all analyses. If mass was saved due to a lighter wing structure, the same amount of mass was accounted for in form of additional payload. In other words, the total mass never changed but the mass and rotational inertia *distributions* changed.

The only *external loads* considered were aerodynamic loads on the main wing and tail. Figure 10 conceptually illustrates how loads from the horizontal tail plane (HTP) were taken into account. Tail plane loads were computed with respect to its orientation, then projected onto the wing centre node and superimposed with the wing loads. Details on how to formulate this mathematically are given in section B-C. Note that tail loads were only included in the manoeuvre analysis.

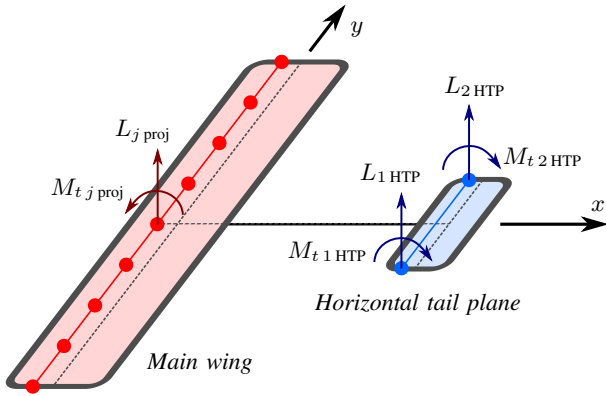


Figure 10: Modelling of the HTP aerodynamic loads by projection of the tail loads onto the wing centre node

The load cases considered for the structural sizing of the wing were pull-up and pull-down manoeuvres at load factors n_z of +4.4 and -1.8. The manoeuvres were assumed to be *quasi-steady* meaning that inertial reaction forces are constant in time and aerodynamic forces are steady. Loads from the HTP including loads from elevator deflections were considered when performing the analysis. Tail and elevator properties are given in section B-A. All manoeuvre calculation were performed for an airspeed of 77 m/s. Details on the mathematical formulation of the pull-up analysis are given in section B-D.

The assessment of stresses resulting from the investigated flight manoeuvres was treated in a simplified manner. As required by CS-23.303 [1], a factor of safety F_S of 1.5 was taken into account. Normal stresses σ in the spar were assessed using the *maximum-principal-stress theory* [9] ($\sigma_y/\sigma_{\text{spar}} > F_S$). Shear stresses in the skin were assessed using the *maximum-shear-stress theory* (Tresca theory, $(\sigma_y/2)/\tau_{\text{skin}} > F_S$), which is a simple and conservative approach for design applications [9]. Normal stresses in the skin were ignored in the stress analysis which is a non-conservative simplification.

C. Structural optimisation

The wing structure was optimised with the objective to achieve minimal structural weight, thus allowing to carry more payload. As indicated in the previous sections this optimisation is constrained by strength and stability requirements. A full list of requirements is given below.

- *Strength constraints*
 - Maximum stresses in quasi-steady +4.4 g pull-up
 - Maximum stresses in quasi-steady -1.8 g pull-down
- *Aerodynamic constraints*
 - Maximum manoeuvre lift coefficient $|C_L| < 1.5$
- *Stability constraints*
 - Lower limit of divergence speed
 - Lower limit of aileron reversal speed
 - Lower limit of flutter speed

The main design variables to achieve a light wing structure were the skin thickness t_s and the spar size A_c (see fig. 9). The skin thickness was set to a constant value of 1.2 mm which is in the range of typical thicknesses for metal skins [10], [11]. The wing spar area A_c was defined as a function of the wing span position. Generally, any arbitrary function may be defined, but in order to keep the number of variables low a second order polynomial was chosen such that the maximum spar area was located at the root.

$$A_c = A_0 - A_1 \left| \frac{y}{l} \right| - A_2 \left| \frac{y}{l} \right|^2 \quad (11)$$

The constant parameters A_0 , A_1 and A_2 define the size of the wing spar and can be considered as design choices. A list of reasonable wing spar designs can easily be generated by a computer (section B-E).

Due to the complex nature of the optimisation with multiple constraints and non-linear problems a brute-force optimisation scheme was chosen. The optimisation procedure is described in more detail in section B-E. The flutter analysis was performed using a modal analysis [7] as this method can significantly reduce computation time.

D. Results

In total, 22 090 wing spar designs based on eq. (11) were analysed. The algorithm produced 28 solutions, where each new solution was a weight improvement compared to the previous one. The wing spar geometry for the final optimised wing structure was $A_c = 119.3 \text{ cm}^2 - 118.7 \text{ cm}^2 |y/l|$.⁷ Figure 11 shows the wing mass for each solution as well as the corresponding minimum factors of safety which were all encountered in the +4.4 g pull-up manoeuvre. In most (but not all) cases the safety factor based on the spar normal stresses $F_{S\sigma}$ was lower than the safety factor based on the skin shear stress $F_{S\tau}$. In the final solution the wing skin stress during the pull-up was the limiting factor of the design.

⁷Most solutions had a non-zero value of A_2 , but the final solution was a linearly tapered spar.

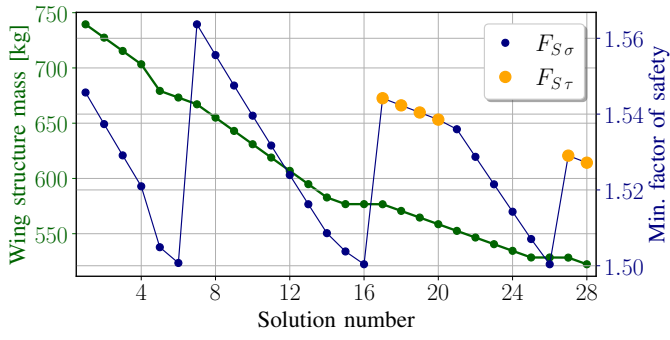


Figure 11: Wing structure mass and minimum factor of safety for each solution produced by the optimisation algorithm

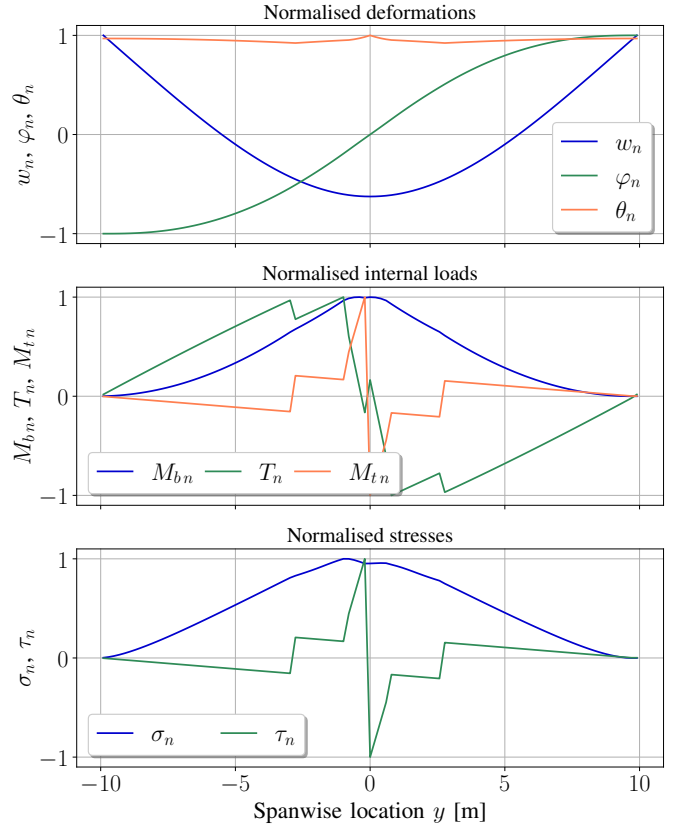
A mass break down of the optimised structure is provided in table III. Note that all masses except the wing structure mass and payload mass were constant for all analyses. Since the stresses at the manoeuvres were found to be the limiting factor for many designs, the chosen theory of elastic failure and the exact stress computation may have significant influence on the exact numerical values of the masses. This is further discussed in section IV-E.

Table III: Mass break down for the optimised DHC-6

Description	Mass	Comment
Fuselage and tail	1710 kg	—
Fuel	1170 kg	Maximum capacity
Engines	660 kg	Left and right engine
Wing equipment	198 kg	—
Wing structure	522.3 kg	} Optimisation results
↳ Wing spar	328.6 kg	
↳ Wing skin	193.7 kg	
Payload	1409.7 kg	
Maximum take-off mass	5670 kg	Defined in CS-23 [1]

Plots of the deformation, internal loads and stresses resulting from the pull-up and pull-down manoeuvres are shown in figs. 12 and 13, respectively. The wing bends and twists due to the manoeuvres. The engines are mounted some distance away from the elastic axis and their inertia clearly introduces both a discrete transverse force (T) and a discrete torsional moment (M_t). In a similar manner, inertial loads from the fuselage can be seen in the centre. The tail loads which were projected onto the wing centre node can also be seen in a change of the transverse force.

Critical speeds for divergence, control reversal and flutter for the final design are summarised in table IV. All computed speeds are in a regime in which effects of compressibility and turbulence (Mach number and Reynolds number effects) should be of concern [13]. The potential flow model does not take these into account so that the validity of these numbers is questionable. Figure 14 illustrates the decrease of aileron effectiveness with increasing airspeed. As the critical airspeed is approached the loss of effectiveness ($\partial C_R / \partial u$) becomes more pronounced, until, at reversal speed, the aileron deflection does not produce any rolling moment ($C_R = 0$).



w_m [m]	φ_m [°]	θ_m [°]	M_{bm} [Nm]	T_m [N]	M_{tm} [Nm]	σ_m [MPa]	τ_m [MPa]
0.39	5.3	10.4	367009	65931	77345	186.6	94.6

Figure 12: Normalised spanwise deformations, internal loads and stresses for a quasi-steady +4.4 g pull-up manoeuvre (trim state: $\alpha = 10.4^\circ$, $\delta_e = -35.0^\circ$ (outside range [12]), $C_L = 1.36$). The table shows the values used to normalise the plots.

Table IV: Critical speeds of the DHC-6 Twin Otter. The never-exceed speed V_{NE} (CS-23.1505 [1]) was assumed to be 103 m/s.

Parameter	Divergence	Reversal	Flutter
u [m/s]	296.5	219.4	314.6
u/V_{NE}	2.88	2.13	3.05

E. Discussion of the analysis

Given the simplifying assumptions of the aeroelasticity model itself and the modelling of the DHC-6, the general trends of the results presented in section IV-D seem reasonable. The final design was limited by the structural strength. The critical speeds due to aeroelastic effects were rather high and have been commented on.

The boundary conditions implemented for the model assumed that the wings were clamped in a single node at the centre of the wing. This might not represent the structural supports very well. The technical specification [8] states that “each mainplane is attached to the fuselage structure, at roof level, by two bolts through fork and lug fittings at the front and rear spars”. An illustration of the structure is also provided by Megson [14, p. 401]. In addition, there are struts supporting the

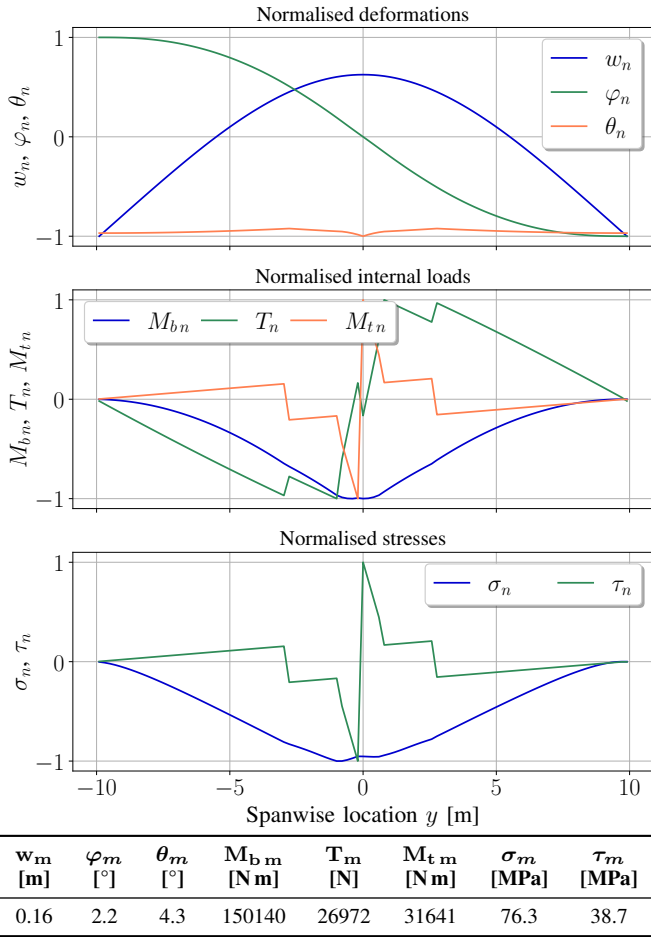


Figure 13: Normalised spanwise deformations, internal loads and stresses for a quasi-steady -1.8 g pull-up manoeuvre (trim state: $\alpha = -4.3^\circ$, $\delta_e = 14.3^\circ$, $C_L = -0.55$). The table shows the values used to normalise the plots.

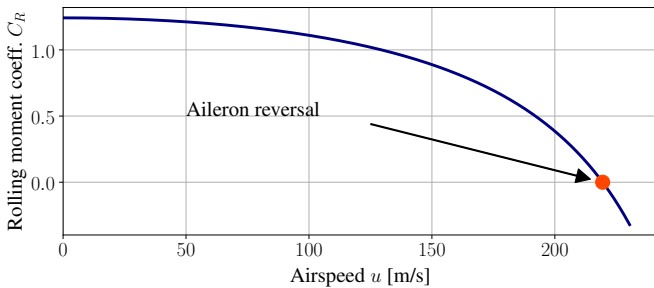


Figure 14: Rolling moment coefficient (aileron deflection $\delta_a = 1$)

wings. These do also impose structural boundary conditions which were not taken into account here. As the struts are mounted at an angle they will not only introduce vertical loads but also spanwise loads. The 3 dof FE model cannot represent these without introducing new degrees of freedom.

It was also observed that the chosen theory of elastic failure had significant influence on the results, especially since strength was a limiting factor for the design. When using *von Mises theory* [9] instead of *Tresca theory* (neglecting skin

normal stresses), a thinner skin thickness could be used as the maximum allowable shear stress was higher.

More accurate modelling of structural boundary conditions might allow lighter structures due to the additional supports. On the other hand, this effect might be somewhat balanced with a better implementation of the elastic failure theory (i. e. taking normal stresses in the skin into account). Of course, there are also several other modes of structural failure like instabilities (e. g. buckling of the skin or the struts [4]) or fatigue problems. These were not treated in the presented analysis but have to be analysed according to the certification specifications [1].

Besides optimising only the wing spar it is of course possible to also include the skin thickness in the optimisation process. The brute-force optimisation scheme certainly has shortcomings as the number of possible designs variations (and thus computation time) increases significantly with each new variable.

V. CONCLUSIONS

This paper described the implementation of a linear aeroelastic model coupling engineering beam theory and a potential flow model in a FE formulation. Predictions of aeroelastic instabilities for a simple wind tunnel model were made and subsequently compared to experiments. The model was then applied to size a simplified wing structure of the DHC-6 Twin Otter. A procedure of how to approach a complex design task with multiple constraints has been demonstrated in section IV.

Even though the model was limited to linear structural and aerodynamic theories as well as to the modelling simplifications, results indicated correct trends. The experiments also revealed some limitations inherent to the underlying aerodynamics model that was used. While strip theory always made conservative predictions, the DLM, being a more accurate model and taking spanwise effects into account, was closer to the experimental observations.

Aeroelastic models like the one presented in this work can be a helpful tool in the preliminary design of aircraft and be used to generate base-line data for more detailed design work. This is of particular interest for novel, unconventional aircraft configurations [3]. For instance, Seywald demonstrated that beam theory and potential flow theories can be used to make mass predictions for box-wing or C-wing configurations [15], but also also to make an early assessment of expected flight dynamics and handling qualities [2].

Possible steps for more accurate modelling of the presented work has been discussed in section IV-E. At the same time limitations of the underlying models (small deformation, potential flow, etc.) must of course be kept in mind.

APPENDIX A

WIND TUNNEL MODEL

A. Model geometry and material parameters

Figure 15 illustrates the general geometry of the wind tunnel model that was used for the experimental verification of the aeroelasticity model (section III). The wing is made from a glass fibre reinforced polymer (GFRP) plate and has a rectangular planform. At the trailing edge there are three independently adjustable aileron sections. At seven spanwise positions there are hinge connections or adjustment rods. The semi-span is $l = 1600$ mm, the semi-chord is $b = 175$ mm and the aileron semi-chord is $b_a = 30$ mm. The plate thickness was determined to be 4.1 mm. The wing was clamped at the wing root when mounted in the wind tunnel.

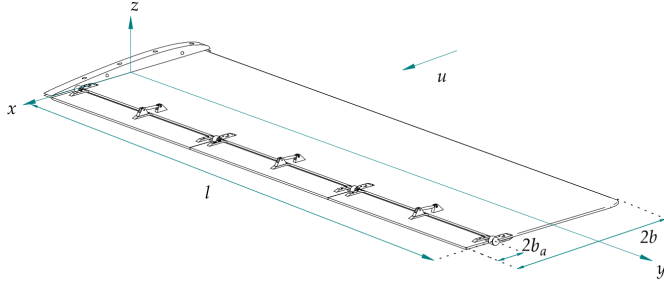


Figure 15: Wind tunnel model geometry

Section III-A explains how the material parameters were determined and calibrated. Table V summarises the Young's modulus E and the shear modulus G of the anisotropic plate in the relevant direction, as well as the plate density ρ and total mass m_{total} used in the computational model.

Table V: Material parameters of the wind tunnel model

E [GPa]	G [GPa]	ρ [kg/m ³]	m_{total} [kg]
32.5	5.1	1960	4.856

APPENDIX B

DHC-6 TWIN OTTER MODEL

A. Geometry, inertia and aerodynamic data

The DHC-6 was modelled as fully symmetric about its roll axis. Figure 16 summarises the most basic geometric model parameters used in the analysis. The distance d_t between the wing reference line (elastic axis) and the tail reference line (quarter-chord) was set to 8.5 m.

The mass, rotational inertia and stiffness properties (EI and GK) of the wing structure were modelled based on the wing spar geometry and the wing box geometry (NACA 63A-516). Additionally, wing equipment was accounted for with a distributed mass and distributed rotational inertia (10 kg/m and 3.5 kgm²/m with respect to the mid-chord line). Engines were modelled as point masses at locations indicated in fig. 16. Each engine mass was 330 kg and its rotational inertia with respect to the engine centre of mass was 250 kgm² about the roll axis and 650 kgm² about the pitch axis. Masses for fuselage, tail, fuel and payload have been stated in table III. Their respective

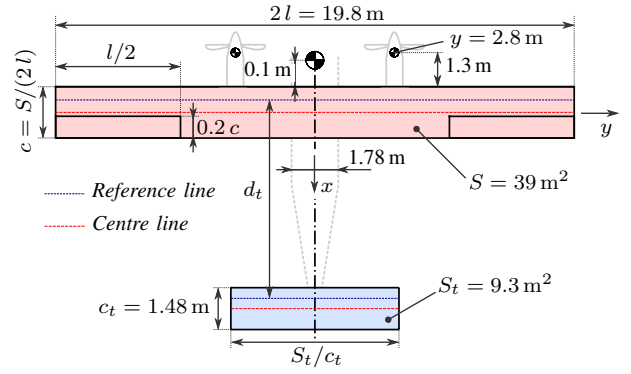


Figure 16: Overview of the DHC-6 Twin Otter model geometry (not to scale)

values for the rotational inertia has been computed as $J = m r_g^2$ where m is the mass and r_g is the radius of gyration which was assumed to be 0.5 m for the roll axis and 1.7 m for the pitch axis for all masses. The inertial data of these components was uniformly distributed over the fuselage width.

Elevator deflections were considered when computing the trim state for the quasi-steady pull-up manoeuvres. The elevator lift coefficient derivative used was $C_{L\delta_e} = \partial C_L / \partial \delta_e = -0.608$ and the pitch moment coefficient derivative $C_{m\delta_e} = \partial C_m / \partial \delta_e = -1.74$ [12].

B. Material parameters

The wing structure of the DHC-6 Twin Otter was assumed to be made fully from A12024-T3. The relevant material parameters are summarised in table VI.

Table VI: A12024-T3 material parameters [16], [17]

E [GPa]	G [GPa]	ρ [kg/m ³]	σ_y [MPa]
72.39	27.58	2768	289

C. Modelling of the horizontal tail plane

Aerodynamic loads from the tail were mapped onto the wing FE model as described in section IV-B and conceptually shown in fig. 10. This section gives a few more mathematical details about this projection.

First, the aerodynamic load vector (cf. eq. (2)) can be divided into a contribution from the wing (subscript w) and a contribution from the tail (subscript t).

$$\mathbf{f}_a = q \mathbf{Q}(k) \mathbf{v} = q (\mathbf{Q}_w + \mathbf{Q}_t) \mathbf{v} = \mathbf{f}_{aw} + \mathbf{f}_{at} \quad (12)$$

Here, $\mathbf{f}_{at}$ is the tail aerodynamic load described in the reference system of the wing FE model. As the HTP was considered to be a rigid structure with a rectangular planform, a single finite element was sufficient to compute loads from the tail. The nodal tail loads (indicated in blue in fig. 10) constitute a 6×1 element load vector $\mathbf{f}_{at\text{elem}}$ in the tail reference system. To obtain the tail load contribution in the wing reference system the vector has to be projected which can be done with a transformation matrix $\mathbf{T}_1$.

$$\mathbf{f}_{at} = \mathbf{T}_1 \mathbf{f}_{at\text{elem}} = \mathbf{T}_1 q \mathbf{Q}_t \mathbf{v}_{t\text{elem}} \quad (13)$$

The tail aerodynamic load matrix $\mathbf{Q}_{t\text{elem}}$ is a 6×6 matrix describing nodal loads as a function of the tail element displacement vector $\mathbf{v}_{t\text{elem}}$. In this case the tail was assumed to be rigidly attached to the wing centre box such that the tail displacement may be described by the displacement of the wing centre node of the FE model using a second transformation.

$$\mathbf{v}_{t\text{elem}} = (w_1, \varphi_1, \theta_1, w_2, \varphi_2, \theta_2)^T_{t\text{elem}} = \mathbf{T}_2 \mathbf{v} \quad (14)$$

Assembling all equations, the tail load contribution can thus simply be added in the following form.

$$\mathbf{f}_a = q \mathbf{Q}(k) \mathbf{v} = q (\mathbf{Q}_w + \mathbf{T}_1 \mathbf{Q}_{t\text{elem}} \mathbf{T}_2) \mathbf{v} \quad (15)$$

Assuming small deformations, the transformation matrices may be written as

$$\mathbf{T}_1 = \begin{bmatrix} \mathbf{0} & & \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -d_t & 0 & 0 & -d_t & 0 & 0 \\ \mathbf{0} & & & & & \end{bmatrix} \quad (16)$$

and

$$\mathbf{T}_2 = \begin{bmatrix} 1 & 0 & -d_t \\ 0 & 0 & 0 \\ 0 & 0 & 1 \\ \mathbf{0} & 1 & 0 & -d_t & \mathbf{0} \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (17)$$

where all zero matrices ($\mathbf{0}$) are of size $(n_{\text{dof}} - 3)/2 \times 2$ (mapping on wing centre node) such that $\mathbf{T}_1$ is of size $n_{\text{dof}} \times 6$ and $\mathbf{T}_2$ of size $6 \times n_{\text{dof}}$ (and n_{dof} being the number of dof in the FE model). Instead of projecting the tail loads onto a single node it is of course also possible to project these onto multiple nodes which might be a more realistic representation of the structural load transfer.

D. Quasi-steady pull-up manoeuvre

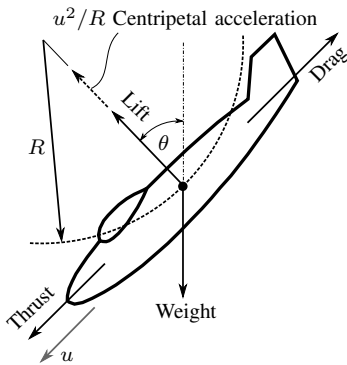


Figure 17: Basic loads in a steady pull-out (adapted from [14])

As mentioned in section IV-B the pull-up and pull-down manoeuvres were treated as quasi-steady manoeuvres which allows to use the steady-state aerodynamic matrix $\mathbf{Q}_0$ (see [7]). Furthermore, small wing deformations were assumed. The trim and wing deformation state were computed for the point of the

maximum load factor (see [14, p.439], fig. 17). The pull-up manoeuvre implies that the rigid body accelerations are known as

$$\ddot{\mathbf{v}}_r = n_z g \mathbf{e}_1 \quad (18)$$

where n_z is the load factor, g the gravitational acceleration and $\mathbf{e}_1 = (1, 0, 0, 1, 0, 0, \dots)^T$ with size $n_{\text{dof}} \times 1$.

The dynamic equilibrium of the elastic wing structure and the aerodynamic and inertial loads acting on this structure can be stated as [7]

$$\underbrace{\mathbf{Z}^T (\mathbf{K} - q \mathbf{Q}_0) \mathbf{Z} \mathbf{v}_p}_{\text{Projected elastic and aerodynamic loads}} - \underbrace{\mathbf{Z}^T (q \mathbf{Q}_0 \mathbf{e}_3) \alpha}_{\text{Projected aerodynamic loads due to } \alpha} = \underbrace{-\mathbf{Z}^T \mathbf{M} \ddot{\mathbf{v}}_r}_{\text{Projected inertial loads}} \quad (19)$$

where $\mathbf{Z}$ is the null space of the matrix of linear constraints $\mathbf{B}$ (for details see [7]). $\mathbf{v}_p$ is the projected nodal displacement vector. The vector $\mathbf{e}_3 = (0, 0, 1, 0, 0, 1, \dots)^T$ (size $n_{\text{dof}} \times 1$) and α is the angle of attack (rigid body orientation). If three dof were structurally constrained, eq. (19) would comprise $n_{\text{dof}} - 3$ equations.

The rigid body pitch orientation is still unknown. However, a second scalar equation from rigid body flight mechanics considerations can be formulated which states that the total lift must equal the vertical inertial loads.

$$\underbrace{\mathbf{e}_1^T q \mathbf{Q}_0 \mathbf{Z} \mathbf{v}_p}_{\text{Lift from elastic wing deformation}} + \underbrace{\mathbf{e}_1^T q \mathbf{Q}_0 \mathbf{e}_3 \alpha}_{\text{Lift due to } \alpha} + \underbrace{q S C_{L\delta_e} \delta_e}_{\text{Lift from elevator deflection}} = \underbrace{\mathbf{e}_1^T \mathbf{M} \ddot{\mathbf{v}}_r}_{\text{Vertical inertia load}} \quad (20)$$

In eq. (20) δ_e is the elevator deflection. A third equation is needed enforcing a pitch moment equilibrium. Pitch moment contributions from the wing and from the tail are being incorporated in this equation as

$$\underbrace{\mathbf{e}_3^T q \mathbf{Q}_0 \mathbf{Z} \mathbf{v}_p}_{\text{Moment from elastic wing deformation}} + \underbrace{\mathbf{e}_3^T q \mathbf{Q}_0 \mathbf{e}_3 \alpha}_{\text{Moment due to } \alpha} + \underbrace{q S 2 b C_{m\delta_e} \delta_e}_{\text{Moment from elevator deflection}} = 0 \quad (21)$$

When setting up these equations one must be careful that all load contributions obey the same coordinate system and the same reference point for the moment equilibrium.⁸ The reference areas for moment and lift coefficients must also be used correctly. The eqs. (19) to (21) form a system of equations which is to be solved for the elastic deformation and trim (α and δ_e). For the solution of this problem it may be convenient to assemble all equations in matrix form with

$$\mathbf{A} := \begin{bmatrix} \mathbf{Z}^T (\mathbf{K} - q \mathbf{Q}_0) \mathbf{Z} & -\mathbf{Z}^T (q \mathbf{Q}_0 \mathbf{e}_3) & \mathbf{0} \\ \mathbf{e}_1^T q \mathbf{Q}_0 \mathbf{Z} & \mathbf{e}_1^T q \mathbf{Q}_0 \mathbf{e}_3 & q S C_{L\delta_e} \\ \mathbf{e}_3^T q \mathbf{Q}_0 \mathbf{Z} & \mathbf{e}_3^T q \mathbf{Q}_0 \mathbf{e}_3 & q S 2 b C_{m\delta_e} \end{bmatrix}$$

⁸Moments originating from $\mathbf{Q}_0$ in eq. (21) are moments with respect to the wing elastic axis, but were here treated as moments around the aircraft centre of mass which is not entirely correct.

so that

$$\begin{pmatrix} \mathbf{v}_p \\ \alpha \\ \delta_e \end{pmatrix} = \mathbf{A}^{-1} \begin{pmatrix} -\mathbf{Z}^T \mathbf{M} \ddot{\mathbf{v}}_r \\ \mathbf{e}_1^T \mathbf{M} \ddot{\mathbf{v}}_r \\ 0 \end{pmatrix} \quad (22)$$

Here, the vector $(\mathbf{v}_p, \alpha, \delta_e)^T$ includes the wing deformation and required angle of attack α and elevator deflection δ_e . To recover the full vector of nodal displacements the projection $\mathbf{v}_p$ is multiplied by the null-space basis, and the wing pitch due to α is superimposed.

$$\mathbf{v} = \mathbf{Z} \mathbf{v}_p + \mathbf{e}_3 \alpha \quad (23)$$

The aerodynamic loading on the wing can simply be computed as $\mathbf{f}_a = q \mathbf{Q}_0 \mathbf{v}$ and the total aircraft lift is the sum of wing and tail lift $L = \mathbf{e}_1^T \mathbf{f}_a + q S C_{L \delta_e} \delta_e$.⁹

E. Optimisation algorithm

This section gives a few more details on the optimisation described in section IV-C. From a list of pre-generated feasible design options each design was checked against the constraints. If a design failed to fulfil a constraint it was discarded. The following pseudocode outlines the practical implementation of this scheme. Note that the flutter analysis was performed last as is required the most computation time.

```
# Pseudocode for the optimisation algorithm
1
2
best_design = base_line_design
3
4
for current_design in design_list:
5
    current_design.initialise()
6
    if current_design.mass.wing >
7
        best_design.mass.wing
8
        continue
9
10
    current_design.divergenceAnalysis()
11
    if current_design.div_speed < min_crit_speed
12
        continue
13
14
    current_design.reversalAnalysis()
15
    if current_design.rev_speed < min_crit_speed
16
        continue
17
18
    current_design.loadAnalysis("4.4g pull-up")
19
    if current_design.stress_safety_factor < 1.5 or
20
        abs(current_design.C_L) > 1.5
21
        continue
22
23
    current_design.loadAnalysis("-1.8g pull-down")
24
    if current_design.stress_safety_factor < 1.5 or
25
        abs(current_design.C_L) > 1.5
26
        continue
27
28
    current_design.flutterAnalysis()
29
    if current_design.flutter_speed < min_crit_speed
30
        continue
31
32
# Design has passed all checks
best_design = current_design
```

In this case the design list was a list of coefficients for the wing spar geometry as described by eq. (11). Reasonable coefficients of this function should be $A_1, A_2, A_3 > 0$ with $A_0 \geq A_1 + A_2$. A list of reasonable coefficients can

simply be generated as shown by the following pseudocode. Values for A_{c_min} and A_{c_max} can easily be obtained from considerations of a minimum allowable spar size and a maximum wing mass.

```
# Pseudocode for generating a list of designs
1
2
Ac_list = []
3
4
for A0 in linspace(Ac_max to Ac_min):
5
    for A1 in linspace(Ac_max to 0):
6
        for A2 in linspace(Ac_max to 0):
7
            if A1 + A2 > Ac_min:
8
                continue
9
            else
10
                Ac_list.append([A0, A1, A2])
11
```

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⁹Note that HTP loads might be included in the $\mathbf{Q}_0$ matrix (see section B-C).